

Fine-Tuning Showdown:

OpenAI vs QLoRA

| A Comparative Analysis of LLM Fine-Tuning Approaches

- Project overview: Comparing proprietary vs open-source fine-tuning
- Dataset: MedQA (12,000 medical board exam questions)
- Goal: Build decision framework for choosing the right approach

The Problem & Motivation

- Generic LLMs struggle significantly with **specialized domains** like medicine and law.
- ChatGPT baseline performance hits a ceiling at **65% accuracy** on medical questions.
- Human benchmarks set a high bar:
 - Medical students: 75-80%
 - Specialists: 85-90%

CORE QUESTION

"Which fine-tuning approach should we use to bridge this gap?"



Results - Performance

METRIC	BASELINE (ChatGPT)	OPENAI (GPT-4o-mini)	QLORA (Llama 3.2-3B)
🎯 Accuracy	65%	85.3%	83.8%
🕒 Latency	-	215ms	850ms
⌚ Training Time	-	25 min	90 min
💰 Training Cost	-	\$18.50	\$1.20



KEY FINDING

Both approaches improved **18-20%** over baseline.
OpenAI is slightly more accurate and faster, while QLoRA is much cheaper.

Methodology & Experiment Design

CORE SETUP

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Dataset: MedQA
12,000 medical board exam questions sourced from USMLE.
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Task: Multiple Choice Q&A
Standardized A/B/C/D format for objective evaluation.

 FAIR COMPARISON PROTOCOL

Both models evaluated on the exact same held-out test set of **1,000 questions** using identical accuracy metrics.

THE CONTENDERS

OpenAI Managed Service	
Model	GPT-4o-mini
Training Data	2,000 examples
QLoRA Self-Hosted	
Model	Llama 3.2-3B
Training Data	8,000 examples

Results - Cost Analysis

🔧 Training Costs

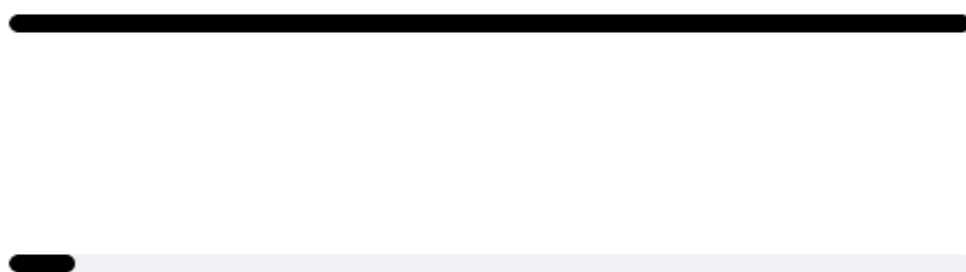
OPENAI

\$18.50

QLORA

\$1.20

93% CHEAPER



☞ Inference Costs (per 1k requests)

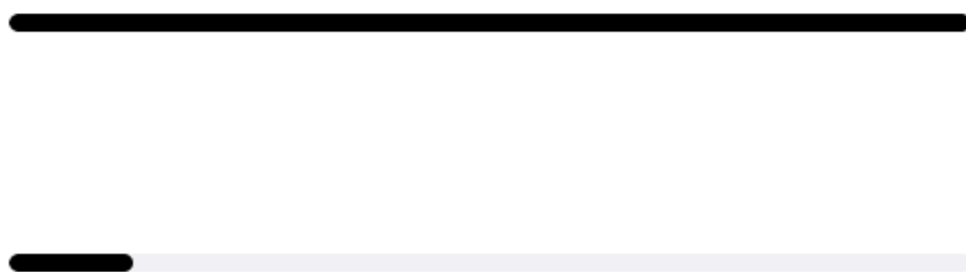
OPENAI

\$0.15

QLORA

\$0.02

87% CHEAPER



BREAK-EVEN POINT



~138,000

Total Inferences



AT 100K / DAY

Pays for itself in 1.4 days



AT 1M / DAY

Pays for itself in 3 hours

Results - Deployment Trade-offs

FACTOR	OPENAI	QLORA
 Time to Deploy	1 day	1 week
 Data Privacy	 Sent to OpenAI	 On-premise
 Scalability	 Auto	 Manual
 Maintenance	None	Ongoing
 Offline Capability	 No	 Yes

Decision Framework

Use OpenAI when...



Need rapid deployment

< 1 week to production

Low inference volume

< 10K requests / day

No dedicated ML infrastructure

Data privacy not critical

Use QLoRA when...



High inference volume

> 100K requests / day

Data privacy is critical

HIPAA, GDPR requirements

Have ML engineering team

Long-term deployment (>6 mos)

RECOMMENDATION

Hybrid Approach

MVP Phase

Start with OpenAI



At Scale

Migrate to QLoRA

Key Takeaways & Recommendations

- 01 No universal winner**
Choice depends heavily on specific constraints like volume and privacy.
- 02 Both approaches work**
Achieved significant 18-20% accuracy improvement over baseline.
- 03 Cost scales differently**
OpenAI scales linearly; QLoRA has flat upfront costs.
- 04 Privacy is the divider**
Healthcare & sensitive data requirements often mandate self-hosted solutions.
- 05 Hybrid approach is viable**
Start fast with managed services, then migrate to optimize costs.

FINAL RECOMMENDATION



Evaluate systematically across performance, cost, and deployment needs. Build a decision framework, not a one-size-fits-all solution.